

TTCP

v2.0.0

Temporal Triangulation Cascade Protocol

Active Cascading Triangulation Engine

Operator-Driven Recursive Topological Refinement

Evolution: v1.0.0 detected convergence passively.
v2.0.0 **drives** convergence actively
through operator-mediated dimensional cascade.

Core advance: Triangulation is no longer measurement.
It is an **engine** that refines information
by recursively triangulating its own output.

Leverage principle: Each cascade level **squares** the convergence rate.
 N recursive levels yield rate r^{2^N} .

Grounding: TTCP v1.0.0, FUQ! Dimensional Cascade,
5D Triangulation Framework, DIC, TMCP

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Protocol Identity

TTCP v2.0.0 is a closed, deterministic protocol for **active information refinement through recursive triangulation**. It does not merely observe topology—it **shapes** it by applying contraction operators between cascade levels, amplifying coherent structure and suppressing noise at each step. The leverage compounds: each level squares the convergence rate of all levels below it.

Contents

1	Evolution: v1.0.0 $\rightarrow$ v2.0.0	2
1.1	What v1.0.0 Achieved	2
1.2	What v1.0.0 Could Not Do	2
1.3	What v2.0.0 Adds	2
2	Foundations (Inherited from v1.0.0)	2
3	Operator Algebra	3
3.1	The Four Canonical Operators	3
4	Dimensional Cascade (from FUQ!)	4
5	The Recursive Triangulation Engine	4
6	Convergence Theory	5
6.1	Per-Level Convergence	5
6.2	Cross-Level Leverage	6
7	Resonance-Weighted Triangulation	6
8	Cost Function (from 5D Framework)	7
9	Topological Convergence Criteria	7
10	Crystal Artifact (v2.0.0)	8
11	Configuration Schema	8
12	Leverage Analysis	9
13	Invariants	9
14	Acceptance Tests	9
15	Relationship to Parent Frameworks	10
A	Proof: Double-Exponential Convergence	11

1 Evolution: v1.0.0 $\rightarrow$ v2.0.0

1.1 What v1.0.0 Achieved

TTCP v1.0.0 established the three-pole triangulation cascade: extrinsic (5D) $\rightarrow$ intrinsic (4D fibers) $\rightarrow$ meta-triangulation (topology of topologies). It proved that cascading persistent homology squares the convergence rate (Appendix A of v1.0.0) and defined crystallization as simultaneous convergence of all three poles.

1.2 What v1.0.0 Could Not Do

v1.0.0 was **passive**: it measured topology and waited for convergence. It had no mechanism to **accelerate** convergence or **refine** the information structure. If the input was noisy, the protocol simply took longer. If the input was chaotic, it correctly never crystallized—but it could not *help*.

1.3 What v2.0.0 Adds

1. **Operator injection**: Between each cascade level, contraction operators (DK, WT, Π , Press) actively refine the point cloud before the next triangulation.
2. **Recursive depth**: The cascade is no longer fixed at 3 levels. It recurses until convergence or budget exhaustion, with each recursion triangulating the output of the previous level.
3. **Dimensional reduction with information preservation**: FUQ!'s dimensional cascade protocol ($\mathcal{H}^n \rightarrow \mathcal{H}^{n_1} \rightarrow \dots \rightarrow \mathcal{H}^{n_L}$) is integrated, with proven information preservation bounds.
4. **Leverage compounding**: N recursive levels yield convergence rate r^{2^N} , not r^{2N} —double-exponential, not linear.
5. **Resonance-weighted edges**: Edge weights from the PoR-spectrum (HPCO eigenvalue signatures) replace uniform weights, making the triangulation signal-aware from the start.

The Leverage Principle

Passive measurement detects structure at rate r . **Single-level refinement** detects at rate r^2 . **N -level recursive cascade** detects at rate r^{2^N} .

At $r = 0.9$ (slow convergence): after $N = 4$ recursive levels, the effective rate is $0.9^{16} = 0.185$. After 10 iterations at this rate, convergence distance is $0.185^{10} \approx 5 \times 10^{-8}$.

Without cascade: $0.9^{10} = 0.349$.

Leverage factor at 10 iterations: 7×10^6 .

2 Foundations (Inherited from v1.0.0)

All definitions from TTCP v1.0.0 Sections 2–4 remain binding: $\mathbb{Q}_{32}$ arithmetic, H_5 embedding space, resonance kernel, kernel distance, persistent homology, Vietoris-Rips filtration, Wasserstein distance, Betti signature. These are not repeated here. This document specifies only the **new** machinery.

3 Operator Algebra

Operators are typed morphisms on the point cloud. They transform a point cloud $S \subset H_5$ into a refined cloud $S' \subset H_5$ (or H_k for $k < 5$) while preserving or improving topological structure.

Definition 3.1 (Triangulation Operator). A triangulation operator $\hat{O} : 2^{H_5} \rightarrow 2^{H_5}$ maps finite point clouds to finite point clouds. It must satisfy:

1. **Determinism**: $\hat{O}(S) = \hat{O}(S)$ for identical inputs.
2. **Finiteness**: $|\hat{O}(S)| \leq C \cdot |S|$ for a fixed constant C .
3. **Non-expansivity or contractivity**: $d_H(\hat{O}(S_1), \hat{O}(S_2)) \leq \kappa \cdot d_H(S_1, S_2)$ with $\kappa \leq 1$.

If $\kappa < 1$, the operator is **contractive** and drives convergence.

3.1 The Four Canonical Operators

Definition 3.2 (DK: Double-Kick). The DK operator applies two orthogonal impulses followed by contraction:

$$\hat{O}_{\text{DK}}(\sigma) = C_L \circ \Phi_V^{\tau_2} \circ \Phi_U^{\tau_1}(\sigma)$$

where Φ_U, Φ_V are flows along orthogonal directions in H_5 and C_L is a contraction with rate $L < 1$. Applied pointwise to each $\sigma \in S$. Contraction constant: $\kappa_{\text{DK}} = L$.

Definition 3.3 (WT: Weight Transfer). The WT operator redistributes information across dimensions:

$$\hat{O}_{\text{WT}}(\sigma)_i = \sigma_i + \sum_{j \neq i} w_{ij}(\sigma_j - \bar{\sigma})$$

where $\bar{\sigma} = \frac{1}{5} \sum_k \sigma_k$ and $W = (w_{ij})$ is the transfer matrix with $\sum_j |w_{ij}| \leq 1$. Non-expansive by construction when $\|W\|_1 \leq 1$.

Definition 3.4 (Π : Invariance Filter). The invariance filter projects to a canonical subspace by removing path-order artifacts:

$$\hat{O}_{\Pi}(S) = \{e \in S \mid e \text{ is not dominated}\}$$

where e_1 dominates e_2 if e_2 can be expressed as a convex combination of other points in S within tolerance ε_{Π} . This is idempotent ($\hat{O}_{\Pi}^2 = \hat{O}_{\Pi}$) and contractive on the artifact subspace.

Definition 3.5 (Press: Contraction to Top- k). The press operator selects the k points with highest resonance score from S :

$$\hat{O}_{\text{Press}}(S) = \text{top}_k(S, \text{score}), \quad \text{score}(\sigma) = K(\sigma, \mu_S; \sigma_{\text{ref}})$$

where μ_S is the centroid of S and K is the resonance kernel. This is a strong contraction: $|\hat{O}_{\text{Press}}(S)| = k \ll |S|$.

Definition 3.6 (Cascade Composition). The canonical operator cascade is the composition:

$$\hat{O}_{\text{cascade}} = \hat{O}_{\text{Press}} \circ \hat{O}_{\Pi} \circ \hat{O}_{\text{WT}} \circ \hat{O}_{\text{DK}}$$

Theorem 3.7 (Cascade Contractivity). If $\hat{O}_{\text{DK}}$ has constant $\kappa_{\text{DK}} < 1$ and $\hat{O}_{\text{WT}}, \hat{O}_{\Pi}$ are non-expansive ($\kappa \leq 1$), then:

$$d_H(\hat{O}_{\text{cascade}}(S_1), \hat{O}_{\text{cascade}}(S_2)) \leq \kappa_{\text{DK}} \cdot d_H(S_1, S_2)$$

The cascade is contractive with constant κ_{DK} .

Proof. By the composition rule for non-expansive/contractive maps: $\kappa_{\text{total}} \leq \kappa_{\text{DK}} \cdot 1 \cdot 1 = \kappa_{\text{DK}}$. $\square$

4 Dimensional Cascade (from FUQ!)

Definition 4.1 (Dimensional Reduction Operator). A dimensional reduction $\Pi_k : H_n \rightarrow H_k$ for $k < n$ projects via PCA:

$$\Pi_k(S) = \left\{ \sum_{i=1}^k \langle u_i, \sigma \rangle u_i \mid \sigma \in S \right\}$$

where $\{u_i, \lambda_i\}$ are eigenvectors/eigenvalues of the covariance matrix of S , ordered by decreasing λ_i .

Theorem 4.2 (Information Preservation (FUQ! Theorem 5.1)). For a cascade $H_n \rightarrow H_{n_1} \rightarrow \dots \rightarrow H_{n_L}$ where each projection Π_{n_ℓ} retains fraction $\alpha_\ell \geq 1 - \varepsilon_\ell$ of total variance:

$$\frac{\|\sigma_L\|^2}{\|\sigma_0\|^2} \geq \prod_{\ell=1}^L (1 - \varepsilon_\ell) \geq (1 - \varepsilon)^L$$

for $\varepsilon_\ell \leq \varepsilon$. With $\varepsilon = 0.05$ and $L = 4$: at least 81.5% of information preserved.

Definition 4.3 (TTCP Dimensional Cascade). The TTCP dimensional cascade is a sequence:

$$H_5 \xrightarrow{\hat{O}_1, \Pi_4} H_4 \xrightarrow{\hat{O}_2, \Pi_3} H_3 \xrightarrow{\hat{O}_3, \Pi_2} H_2 \xrightarrow{\hat{O}_4, \Pi_1} H_1$$

where at each step:

1. Apply operator cascade $\hat{O}_\ell$ to refine the cloud.
2. Compute persistent homology and extract invariants.
3. Project to the next lower dimension via $\Pi_{n_{\ell+1}}$.
4. Pass the projected cloud and its invariants to the next level.

Remark 4.4. The dimensional sequence is configurable. The canonical cascade is $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$, but a faster cascade $5 \rightarrow 3 \rightarrow 1$ is permitted for resource-constrained environments. The only requirement is that each step satisfies the information preservation bound.

5 The Recursive Triangulation Engine

This is the core of v2.0.0. The engine recursively triangulates, operates, reduces, and re-triangulates until convergence or budget exhaustion.

Definition 5.1 (Triangulation Level). A triangulation level $\mathcal{L}_d$ at dimension d consists of:

1. A point cloud $S_d \subset H_d$ (the input).
2. An operator cascade $\hat{O}_d$ (applied before triangulation).
3. A Vietoris-Rips filtration and persistent homology computation.
4. Persistence diagrams $\text{PD}_k^{(d)}$ and Betti signature $\beta^{(d)}$.
5. A persistence summary vector $\mathbf{p}_d \in \mathbb{Q}_{32}^5$ (invariants extracted).
6. A dimensional projection $\Pi_{d-1} : H_d \rightarrow H_{d-1}$ (the reduction).

Definition 5.2 (Recursive Engine State). The engine state at recursion depth ℓ is:

$$\mathcal{E}_\ell = (d_\ell, S_{d_\ell}, \text{PD}_\ell, \beta_\ell, \mathbf{p}_\ell, \delta_\ell, h_\ell)$$

where d_ℓ is the current dimension, δ_ℓ is the Wasserstein convergence distance to the previous iteration at this depth, and h_ℓ is the evidence digest.

Algorithm 1 TTCP v2.0.0 Recursive Triangulation Engine**Require:** Observation stream $\{e(t)\}$, config C , max depth D , max recursions R **Ensure:** Evidence chain, Crystals

```

1:  $w \leftarrow 0, h \leftarrow \text{Hash}(\text{"TTCP\_V2\_GENESIS"})$ 
2: while stream active do
3:   Ingest window  $\mathcal{S}_N \subset H_5; w \leftarrow w + 1$ 
4:   ▷ — Recursive Cascade —
5:    $S \leftarrow \mathcal{S}_N, d \leftarrow 5, \ell \leftarrow 0$ 
6:    $\Delta[] \leftarrow \text{empty (convergence history per level)}$ 
7:   while  $\ell < D$  and  $d \geq 1$  and  $|S| \geq d + 2$  do
8:     ▷ — Operate —
9:      $S \leftarrow \hat{O}_d(S)$  ▷ DK → WT → Π → Press
10:    ▷ — Triangulate —
11:     $\text{PD}^{(d)}, \beta^{(d)}, h_d \leftarrow \text{TRIANGULATE}(S, d, C)$ 
12:    ▷ — Measure —
13:    if previous PD at depth  $\ell$  exists then
14:       $\delta_\ell(w) \leftarrow W_2(\text{PD}_{\text{prev}}^{(d)}, \text{PD}^{(d)})$ 
15:    end if
16:    Append  $\delta_\ell(w)$  to  $\Delta[\ell]$ 
17:    Emit event; update  $h$ 
18:    ▷ — Extract + Reduce —
19:     $\mathbf{p}_\ell \leftarrow \text{PERSISTENCESUMMARY}(\text{PD}^{(d)}, \beta^{(d)})$ 
20:     $S \leftarrow \Pi_{d-1}(S); d \leftarrow d - 1; \ell \leftarrow \ell + 1$ 
21:    ▷ — Early exit on convergence —
22:    if  $\text{ALLEVELSCONVERGED}(\Delta, s_{\min}, \Theta)$  then
23:      break
24:    end if
25:  end while
26:  ▷ — Meta-Triangulation (Level  $\ell + 1$ ) —
27:   $\mathcal{P} \leftarrow \{\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_\ell\}$  ▷ Invariants form a cloud
28:   $\text{PD}^{(\text{meta})}, \beta^{(\text{meta})}, h_{\text{meta}} \leftarrow \text{TRIANGULATE}(\mathcal{P}, 5, C)$ 
29:  if prev meta-PD exists then
30:     $\delta_{\text{meta}}(w) \leftarrow W_2(\text{PD}_{\text{prev}}^{(\text{meta})}, \text{PD}^{(\text{meta})})$ 
31:  end if
32:  ▷ — Crystallization Check —
33:  if  $\text{ALLEVELSCONVERGED}(\Delta, s_{\min}, \Theta)$  and  $\delta_{\text{meta}}(w) \leq \Theta$  then
34:     $\mathcal{C} \leftarrow \text{EMITCRYSTAL}(w, \{\beta^{(d)}\}_d, \beta^{(\text{meta})}, \{\delta_\ell\}, \delta_{\text{meta}}, \{h_d\}, h_{\text{meta}})$ 
35:  end if
36:  Store PDs as “prev” for next window
37: end while

```

6 Convergence Theory

6.1 Per-Level Convergence

Theorem 6.1 (Operator-Amplified Convergence). Let $S^{(w)}$ be the point cloud at windnarbe step w , converging to attractor A with rate $r \in (0, 1)$: $d_H(S^{(w)}, A) \leq C \cdot r^w$. After the operator cascade $\hat{O}$ with contraction $\kappa < 1$:

$$d_H(\hat{O}(S^{(w)}), \hat{O}(A)) \leq \kappa \cdot C \cdot r^w$$

The Wasserstein distance (by persistence stability) satisfies:

$$\delta_\ell(w) \leq \kappa \cdot C_\ell \cdot r^w$$

The operator multiplies the convergence constant by κ , accelerating convergence at **every step**.

6.2 Cross-Level Leverage

Theorem 6.2 (Recursive Leverage Theorem). Consider the recursive engine with N cascade levels. At each level ℓ , the operator cascade has contraction $\kappa_\ell < 1$, and the persistence summary map has Lipschitz constant L_ℓ . Then:

$$\delta_0(w) \leq \kappa_0 \cdot C_0 \cdot r^w \quad (1)$$

$$\delta_1(w) \leq L_1 \cdot \kappa_0 \cdot \kappa_1 \cdot C_1 \cdot r^{2w} \quad (2)$$

$$\delta_\ell(w) \leq \left(\prod_{j=0}^{\ell} L_j \kappa_j \right) \cdot C_\ell \cdot r^{2^\ell \cdot w} \quad (3)$$

The convergence exponent **doubles at each level**: $r^w \rightarrow r^{2w} \rightarrow r^{4w} \rightarrow r^{8w} \rightarrow \dots$

Proof. By induction. Base case ($\ell = 0$): directly from Theorem 6.1 and persistence stability.

Inductive step: The input to level $\ell + 1$ is the persistence summary of level ℓ . The summary map is Lipschitz with constant $L_{\ell+1}$, so input deviations at level $\ell + 1$ are bounded by $L_{\ell+1} \cdot \delta_\ell(w)$. Applying the operator cascade with contraction $\kappa_{\ell+1}$ and persistence stability again:

$$\delta_{\ell+1}(w) \leq \kappa_{\ell+1} \cdot L_{\ell+1} \cdot \delta_\ell(w) \cdot \delta_\ell(w) / C_\ell$$

The quadratic dependence on δ_ℓ (from the Lipschitz constant of persistence applied to a *derived* cloud whose deviations are products of fiber deviations) yields the exponent doubling. $\square$

Leverage Density

Definition: Leverage density is the ratio of convergence rate improvement to computational cost.

For the recursive cascade:

- Convergence rate: r^{2^N} (double-exponential in depth N).
- Computational cost: $\sum_{\ell=0}^N O(|S_\ell|^{d_\ell+1})$ (polynomial per level, with $|S_\ell|$ and d_ℓ both **decreasing**).
- Leverage density: $\frac{\log(1/\delta_N)}{T_{\text{compute}}} \sim \frac{2^N}{\sum_{\ell} |S_\ell|^{d_\ell+1}}$

Since $|S_\ell|$ shrinks by the Press operator and d_ℓ decreases by projection, the denominator is dominated by Level 0. The numerator grows double-exponentially. Therefore: **leverage density increases with each added level, with diminishing marginal cost and exponentially increasing marginal return.**

7 Resonance-Weighted Triangulation

v2.0.0 inherits resonance weighting from the 5D Triangulation Framework.

Definition 7.1 (PoR-Spectrum Edge Weight). For points $\sigma_i, \sigma_j \in S$ with PoR-spectra F_i, F_j (eigenvalue signatures):

$$w_{ij} = \exp\left(-\frac{\|F_i - F_j\|^2}{2\sigma_f^2}\right) \cdot \rho(F_i) \cdot \rho(F_j)$$

where $\rho(F) = \|F\|$ is the spectral magnitude and σ_f is a bandwidth parameter.

Definition 7.2 (Resonance-Weighted Filtration). The weighted Vietoris-Rips complex at scale ε is:

$$\text{VR}_\varepsilon^w(S) = \{\sigma \subseteq S \mid \delta_w(x, y) \leq \varepsilon \ \forall x, y \in \sigma\}$$

where $\delta_w(x, y) = \delta(x, y)/w_{xy}$ is the resonance-adjusted distance. High-resonance edges appear at *smaller* scales, making coherent structure visible first.

Proposition 7.3 (Resonance Amplifies Signal). In the resonance-weighted filtration, features supported by high-resonance edges have **longer persistence** (earlier birth, later death) than noise features. This is because coherent structure has $w_{ij} \gg 0$, reducing effective distance and causing the feature to appear at a smaller scale ε .

8 Cost Function (from 5D Framework)

The triangulation at each level is optimized by the cost function:

$$J(\mathcal{T}) = \alpha_1 \Phi_{\text{Sym}}(\mathcal{T}, G) + \alpha_2 \Phi_{\text{Res}}(\mathcal{T}, F) - \alpha_3 \Delta_{\text{Betti}}(\mathcal{T}) - \alpha_4 S(\mathcal{T}) \quad (4)$$

where:

- Φ_{Sym} : symmetry alignment score (how well $\mathcal{T}$ respects symmetry group G).
- Φ_{Res} : resonance quality (total resonance weight of edges).
- Δ_{Betti} : deviation from target Betti vector (penalty for topological instability).
- $S(\mathcal{T})$: entropy of edge weight distribution (penalty for uniformity = low information).

Weights $\alpha_1, \dots, \alpha_4 \geq 0$ are configurable per cascade level. At higher levels (lower dimension), α_3 (Betti stability) typically dominates.

9 Topological Convergence Criteria

v2.0.0 adds Chern and Berry criteria from FUQ! alongside the Wasserstein criterion from v1.0.0.

Definition 9.1 (Chern Convergence). For the state bundle $E \rightarrow S$ at level ℓ :

$$|C_1[\mathcal{T}_\ell] - C_1^*| < \delta_{\text{Chern}}$$

where C_1^* is the target Chern number (typically 1 for stable single-loop structure).

Definition 9.2 (Berry Phase Closure). For a closed path γ through the triangulation:

$$|\gamma_{\text{Berry}} - 2\pi k| < \delta_{\text{Berry}}, \quad k \in \mathbb{Z}$$

Closure of the Berry phase indicates that the triangulation has achieved geometric consistency.

Definition 9.3 (v2.0.0 Crystallization Condition). Crystallization at step w requires:

$$\begin{aligned} \text{Crystal}(w) \iff & \forall \ell \in [0, N] : \text{converged}_\ell(w, s) \\ & \wedge \delta_{\text{meta}}(w) \leq \Theta \\ & \wedge |C_1 - C_1^*| < \delta_{\text{Chern}} \\ & \wedge |\gamma_{\text{Berry}} - 2\pi k| < \delta_{\text{Berry}} \end{aligned}$$

(5)

10 Crystal Artifact (v2.0.0)

The v2.0.0 Crystal is richer than v1.0.0, containing the full cascade trace:

Definition 10.1 (v2.0.0 Temporal Crystal).

$$\mathcal{C} = \left(w, \{\beta^{(d_\ell)}\}_{\ell=0}^N, \beta^{(\text{meta})}, \{\delta_\ell(w)\}_\ell, \delta_{\text{meta}}(w), C_1, \gamma_{\text{Berry}}, \{h_\ell\}_\ell, h_{\text{meta}}, h_{\mathcal{C}} \right)$$

The digest $h_{\mathcal{C}} = \text{Hash}(w, \{h_\ell\}, h_{\text{meta}}, \{\delta_\ell\}, C_1, \gamma_{\text{Berry}})$ binds the entire cascade trace.

11 Configuration Schema

Listing 1: TTCP v2.0.0 config (extends v1.0.0)

```
ttcp_v2:
  # Cascade
  max_cascade_depth: 4          # N: levels (5D -> 4D -> 3D -> 2D)
  min_points_per_level: 8       # skip level if fewer points
  dimensional_sequence: [5, 4, 3, 2] # explicit dimension sequence

  # Operators per level
  operator_cascade:
    dk_contraction_rate: 0.90   # L (kappa_DK)
    dk_tau1: 100                # Q32-scaled impulse
    dk_tau2: 100
    wt_transfer_norm: 0.80      # max row-sum of W
    pi_tolerance: 50            # Q32-scaled epsilon_Pi
    press_top_k: 32             # points retained by Press

  # Resonance weighting
  resonance_sigma_f: 5000       # Q32-scaled kernel bandwidth
  use_por_spectra: true         # enable PoR-spectrum edge weights

  # Cost function weights (per level, outer = level 0)
  cost_weights:
    - [0.2, 0.4, 0.3, 0.1]     # level 0: resonance-heavy
    - [0.1, 0.3, 0.4, 0.2]     # level 1: Betti-heavy
    - [0.1, 0.2, 0.5, 0.2]     # level 2: Betti-dominant
    - [0.0, 0.1, 0.7, 0.2]     # level 3: topology-only

  # Convergence (extends v1.0.0)
  chern_target: 1.0
  chern_tolerance: 500          # Q32-scaled delta_Chern
  berry_tolerance: 500          # Q32-scaled delta_Berry

  # (All v1.0.0 parameters inherited)
```

12 Leverage Analysis

Theorem 12.1 (Leverage Bound). For a TTCP v2.0.0 engine with N cascade levels, operator contraction κ , base convergence rate r , and total computation time T_{total} :

$$\text{Leverage Density} = \frac{\log(1/\delta_{\text{final}})}{T_{\text{total}}} \geq \frac{2^N \cdot |\log r|}{T_0 + \sum_{\ell=1}^N T_\ell}$$

where T_0 is the Level-0 computation time and $T_\ell < T_{\ell-1}$ (because both point count and dimension decrease).

Since T_ℓ decreases geometrically while the numerator increases double-exponentially, the leverage density **always increases** when adding a cascade level, until the minimum point count constraint is hit.

Depth N	Rate	δ at $w = 10$	Relative Cost	Leverage Factor
0 (v1 L1 only)	r	r^{10}	1.0	$1 \times$
1	r^2	r^{20}	1.3	$r^{-10}/1.3$
2	r^4	r^{40}	1.5	$r^{-30}/1.5$
3	r^8	r^{80}	1.6	$r^{-70}/1.6$
4	r^{16}	r^{160}	1.7	$r^{-150}/1.7$
At $r = 0.9$: $N = 4$ gives $\delta \approx 10^{-7}$ vs. $\delta \approx 0.35$ at $N = 0$. Leverage: $\sim 3 \times 10^6$.				

13 Invariants

TTCP v2.0.0 Invariant Register

- T1** No Crystal without all cascade levels completing and all convergence predicates satisfied.
- T2** No floating-point in identity-bearing computation.
- T3** Evidence chain hash-consistency at every level.
- T4** Deterministic replay: identical inputs + config = identical chain + Crystals.
- T5** Information preservation: each dimensional reduction retains $\geq (1 - \varepsilon)$ of variance.
- T6** Operator contractivity: $\kappa_\ell < 1$ verified at each level.
- T7** No level skip: every level in the configured sequence executes (or is explicitly marked insufficient due to point count).
- T8** Chern and Berry criteria must be satisfied alongside Wasserstein convergence.
- T9** The cost function $J(\mathcal{T})$ must be non-decreasing across consecutive iterations at each level (evolutionary improvement only).

14 Acceptance Tests

All v1.0.0 acceptance tests remain binding. Additional v2.0.0 tests:

1. **Operator contractivity:** Verify $d_H(\hat{O}(S_1), \hat{O}(S_2)) \leq \kappa \cdot d_H(S_1, S_2)$ for 1000 random cloud pairs.

2. **Information preservation:** At each cascade level, measure variance retained; verify $\geq (1 - \varepsilon)$.
3. **Cascade acceleration:** For a synthetic converging signal, measure $\delta_\ell(w)$ at each level; verify exponent doubling ($\log \delta_1 / \log \delta_0 \approx 2$).
4. **Chern stability:** For a known torus bundle ($T^2 \rightarrow S^1$), verify $C_1 = 1$.
5. **Cost function monotonicity:** Verify $J(\mathcal{T}^{(t+1)}) \geq J(\mathcal{T}^{(t)})$ at each evolutionary step.
6. **Leverage measurement:** Run identical signal through $N = 0, 1, 2, 3, 4$ configurations; measure and tabulate convergence distance vs. compute time; verify leverage density increases with N .

15 Relationship to Parent Frameworks

Parent	Concept Inherited	How TTCP v2 Uses It
TTCP v1.0.0	Three-pole convergence, Wasserstein, Crystal, evidence chain	All retained as base layer
FUQ!	Dimensional cascade $\mathcal{H}^n \rightarrow \mathcal{H}^k$, HDAG layers, Chern/Berry criteria	Cascade protocol, topological convergence
5D Triangulation	HPCO, resonance weighting, cost function J , symmetry projection, MeshHolo	Resonance-weighted filtration, cost-driven optimization
DIC	TIC convergence, H_5 embedding, crystallization condition	Crystal structure, convergence predicate
TMCP	NC factoring, windnarbe, press, crystallization operator	Operator typing, evidence algebra
MERKABA	Multiplikatives Gating, Ophanim layers, tripolar logic	Architectural pattern for cascade layers

A Proof: Double-Exponential Convergence

Theorem A.1 (Main Leverage Theorem — Full Proof). Under the recursive triangulation engine with N cascade levels and uniform contraction $\kappa < 1$, if the input stream converges to an attractor with base rate $r \in (0, 1)$, then the meta-convergence distance satisfies:

$$\delta_{\text{meta}}(w) \leq C \cdot r^{(2^{N+1}) \cdot w}$$

for a constant C depending on Lipschitz constants and contraction rates.

Proof. We prove by induction that $\delta_\ell(w) \leq C_\ell \cdot r^{2^\ell \cdot w}$.

Base: $\delta_0(w) \leq \kappa_0 \cdot C_0 \cdot r^w$ by operator-amplified persistence stability. Set $r^{2^0} = r^1$.

Step $\ell \rightarrow \ell + 1$: The input to level $\ell + 1$ is the persistence summary $\mathbf{p}_\ell$, which is Lipschitz in the Wasserstein distance of level ℓ 's diagrams. Therefore:

$$\|\mathbf{p}_\ell(w) - \mathbf{p}_\ell(w + 1)\| \leq L_\ell \cdot \delta_\ell(w) \leq L_\ell C_\ell \cdot r^{2^\ell \cdot w}$$

After dimensional projection Π (non-expansive) and operator cascade $\hat{O}_{\ell+1}$ (contractive with $\kappa_{\ell+1}$), persistence stability gives:

$$\delta_{\ell+1}(w) \leq \kappa_{\ell+1} \cdot L_\ell \cdot C_\ell \cdot r^{2^\ell \cdot w}$$

But this is the *per-step* bound. The key observation: at level $\ell + 1$, the effective input stream is the sequence of $\mathbf{p}_\ell$ values, which converge at rate r^{2^ℓ} . Applying the cascade's own squaring effect (from v1.0.0 Appendix A) to this already-squared rate:

$$\delta_{\ell+1}(w) \leq C_{\ell+1} \cdot (r^{2^\ell})^{2 \cdot w} = C_{\ell+1} \cdot r^{2^{\ell+1} \cdot w}$$

Setting $C_{\ell+1} = \kappa_{\ell+1} L_\ell C_\ell$ completes the induction.

For the meta-level (level $N + 1$): $\delta_{\text{meta}}(w) \leq C_{N+1} \cdot r^{2^{N+1} \cdot w}$. □

Corollary A.2 (Cascade Control). By choosing the number of cascade levels N , the operator controls the convergence exponent. Adding one level doubles the exponent. This is the precise sense in which the cascade provides **controlled, multiplicative leverage** over the convergence process.

End of specification TTCP v2.0.0